

Disk & Storage Cheat Sheet

See how full your disks are, find what's using the space, and mount drives at boot.

Using it

Command	Shows	Command	Shows
df -h	Free space per filesystem	lsblk	Disks and partitions (tree)
df -i	Free inodes (file slots) instead	lsblk -f	Plus type, UUID, use%
du -sh /var/log	Total size of one folder	blkid /dev/sdb1	UUID and type
du -h --max-depth=1 sort -h	Each subfolder, biggest last	findmnt	What's mounted where

Reading df -h

Filesystem	Size	Used	Avail	Use%	Mounted on
tmpfs	785M	1.3M	784M	1%	/run ← tmpfs = in RAM; ignore
/dev/sda2	98G	71G	23G	76%	/ ← the root filesystem
/dev/mapper/vg0-data	492G	412G	55G	89%	/srv ← getting full: act above ~90%

Used + Avail < Size: ext4 keeps ~5% for root. **Use% 100%** in **df -i** = out of inodes (too many tiny files).

Reading lsblk

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
sda	8:0	0	100G	0	disk	← a whole disk
-sda1	8:1	0	512M	0	part	/boot/efi ← partitions of sda
`-sda2	8:2	0	99.5G	0	part	/
sdb	8:16	0	500G	0	disk	
`-vg0-data	253:0	0	500G	0	lvm	/srv ← LVM volume built on sdb
nvme0n1	259:0	0	931.5G	0	disk	← no partitions, not mounted: a new, empty disk

TYPE: disk / part (partition) / lvm / rom · **RM 1** = removable · **RO 1** = read-only · no **MOUNTPOINTS** = not mounted

Mounting

Command	Does	Command	Does
mount /dev/sdb1 /mnt	Mounts a partition on a folder	mount -a	Mounts all of fstab (a good test)
umount /mnt	Unmounts it (note: no "n")	findmnt --verify	Checks fstab for mistakes
mount -o remount,rw /	Changes options, stays mounted	lsof +D /mnt	Who is using it ("target is busy")

Reading an /etc/fstab line

# device	mount point	type	options	dump	pass
UUID=3f6c1a2e-8b7d-4c1e-9a55-0d2b7e6f1c90	/srv	ext4	defaults,nofail	0	2

Field	Means	Field	Means
UUID=...	Which partition (UUID from blkid)	options	nofail = boot anyway if disk is missing
/srv	The folder it appears in (must exist)	dump	Old backup flag: always 0
ext4	Filesystem type (xfs , vfat , nfs ...)	pass	Check order at boot: 1 = /, 2 = others, 0 = skip

"Disk full" checklist

Step	Why
df -h then df -i	Which filesystem is full, and is it space or inodes?
sudo du -xh --max-depth=1 / sort -h	Biggest folders (-x: this filesystem only); repeat inside the top one
sudo journalctl --vacuum-size=500M	Old logs. Also try apt clean , old backups, docker system df
sudo lsof +L1	Deleted files still open (df counts them, du can't): restart that process

Remember: after editing **/etc/fstab**, run **sudo mount -a** before rebooting. A bad line can stop the server booting; **nofail** keeps a missing data disk from doing that.